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Evaluating Porosity and Density Measurement of Organic-Rich Tight Rocks

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Abstract

Objective/Scope: Porosity and density of organic-rich source rocks are fundamental reservoir parameters and have been measured using multiple methods. This study seeks to verify the accuracy and advantages of these methods and to determine optimal technology for characterization of source rock reservoirs.

Methods, Procedures, Process: In recent years, combination of nuclear magnetic resonance and gas-porosimetry (CNG), which is a noninvasive and much faster method than existing alternatives, has been established as a standard method to evaluate organic-rich tight rocks. 20 plugs from multiple wells of a source rock reservoir were used to test the accuracy of CNG as a comparison to the industry standard GRI method. The plugs were first measured using CNG, then saturated using light oil at 4000 psig for a week to measure porosity and density. GRI measurements were also performed on crushed samples acquired around the plug locations using the core carcass.

Results, Observations, Conclusions: Porosity and density from three methods are analyzed for convergent validity. The porosities and densities of the 20 samples are very consistent whether using CNG or plug saturation methods. In contrast, significant scattering was observed between these two methods and GRI. By analyzing the sample, experimental procedure, and results, we conclude all the methods are accurate. The observed differences were due to the variations in the samples used for different methods. However, CNG is noninvasive and can be done in hours, in contrast to GRI which is destructive to the sample and takes several weeks to provide needed results.

Novel/Additive Information: CNG is noninvasive and rapid, holding a significant advantage over industry standard GRI method. This study further establishes CNG as a reliable and optimal method to determine petrophysical parameters of organic-rich source rock reservoirs.

Introduction

Nuclear magnetic resonance (NMR) has proven to be extremely valuable in characterizing unconventional source rock reservoirs, both subsurface and in the lab, providing data on porosity, saturation, densities, etc (1, 2). Despite numerous studies, however, the complicated interactions between fluids and rock matrix, particularly organic matter, complicates NMR usage and interpretation of the measured results for source rocks (3) (4) (5).

Currently, the industry standard method to determine porosity, density, and fluid saturation for unconventional source rock reservoirs is GRI, developed at Gas Research Institute for application to Devonian shales using crushed rocks by Luffel and Guidry (6). Because source rock permeability is on the nD scale, even for crushed rocks, it takes weeks to extract the in-situ fluids and subsequently dry the sample for property measurement. Despite the difficulty and length of experiment, GRI remains the most widely used method for assessing fundamental properties of organic-rich source rocks in the petroleum industry.

We have developed and demonstrated a noninvasive and rapid measurement method, combination of NMR and gas-porosimetry (simplified as CNG), to measure source rock porosity and density using intact ‘as received’ plugs (2). The method was investigated and verified by comparing to GRI results using a small set of source rock samples. Significant variations in measured results have been noted for certain samples.

Here we conducted a systematic examination and analysis using more than 200 core samples from multiple wells from a single reservoir by comparing to results from GRI. To further evaluate the CNG method, we extended the core sample testing workflow by pressure saturating a subset of plugs with light oil to determine the porosity and density of the samples. This enables a more objective evaluation of CNG accuracy, as both methods now use the same samples.

An analysis of the testing workflow and measured results indicates that while comparing trends of measured data from CNG and GRI is beneficial, direct comparison on any single sample is not meaningful because samples used for the two methods are not the same. Due to the specificity of plug location compared to the averaging nature of a crushed sample from a larger depth range, small-scale reservoir heterogeneity and sample size/volume difference between samples used for GRI and CNG methods cause significant inconsistency in measured results for associated samples between the two methods.

This study demonstrates that CNG is an optimal laboratory measurement technology for assessing the petrophysical properties of source rocks, offering significant advantages over the industry-standard GRI method. It supports the full deployment of CNG as a standard method for characterizing unconventional source rock reservoirs and suggests its broader application in the petroleum industry.

Method

CNG method

CNG, combination of NMR and gas-porosimetry, was described in reference (2). Briefly, for an ‘as received’ core plug, the total pore volume V_{por} contains liquid filled volume V_l, including both brine and hydrocarbon, and gas-filled void space V_g, with

$$V_{por} = V_g + V_l. \tag{1}$$

A simple NMR experiment measures all the liquid volume in a source rock sample (3, 7, 8). Similarly a Helium gas-porosimetry experiment (9) measures the total gas volume in the plug. The summation of the two measurements is then the total pore volume of a rock sample as in Eq. (1). Note both methods work for a cylindrical plug and are non-damaging to the sample. The bulk volume of a cylindrical core plug V_b can be readily measured from its diameter and length. The porosity is then easily obtained from the CNG method,

$$\phi = \frac{V_g + V_l}{V_b} \tag{2}$$

Measurement of the bulk density of a core plug in an "as received" state seems straightforward, dividing its mass m by bulk volume V_b However, this method does not provide the true bulk density at reservoir condition due to fluid loss during core extraction from subsurface, transportation to the laboratory, and plugging process. In subsurface, the capillary condensation effect renders fluid in nanopores in liquid state for a majority of nanoporous source rocks (3, 10). As a result, we estimated the bulk density in the CNG method by replacing the measured gas volume from gas-porosimetry with a light oil of density $\rho_{oil} = 0.8$ g/ml, in the absence of knowing the true density of the subsurface fluid. Therefore, the bulk density is

$$\rho_b = \frac{m + V_g \rho_{oil}}{V_b} \tag{3}$$

This will allow more effective comparison of measured bulk density from plugs with subsurface logs. It also allows direct comparison of measured results between CNG and saturated plugs in our study.

Grain density of the plug is also obtained from the measurements

$$\rho_g = \frac{m - \rho_l V_l}{V_b - V_l - V_g} \tag{4}$$

Experiments

Test workflow and sampling. A total of 224 samples from cores acquired from multiple wells within one reservoir were used for this study. Fig. 1 a shows the testing workflow. One plug and one associated GRI sample were obtained from each of the 224 core sections, as illustrated in Fig. 1 b.

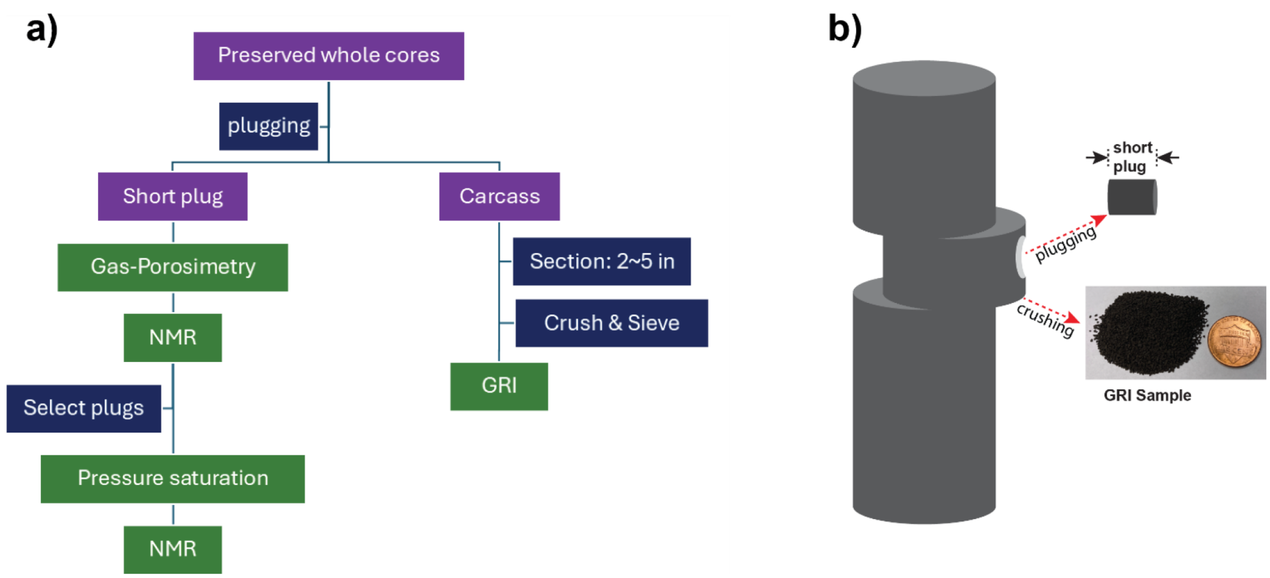

Figure 1—a) Testing workflow for preserved whole cores for porosity and density. The purple, blue, and green blocks represent samples, operations, and data measurements, respectively. b) illustration of taking plug and GRI samples from whole core.

In the measurement workflow, the plugging, gas-porosimitry, and GRI measurement were carried out by external commercial laboratories. All the drilled "short plugs", typically 1" in diameter (with some exception 1.5" diameter) and 1-2" in length, were trimmed into right cylinders. Although it was requested of the commercial lab that GRI sampling make use of the same whole-core section from which the plugs were drilled, as illustrated in Fig. 1b; some GRI samples were chosen from depth ranges above or below the plug depth in the carcass after plugging. Additionally, the GRI section should span approximately two inches of depth centered the plug; but for some samples up to a 6 inch whole core sections were used for GRI by the commercial laboratory. One possible reason is that the particular rock around the plug depth did not yield enough crushed sample within the necessary mesh size for GRI, which is 20 – 35 mesh (0.5-0.84 mm). If the rock tends to break down into pieces smaller than this size range, then more rock would need to be crushed to increase total weight of usable sample for GRI testing.

Pressure saturation of selected plugs. Among the 224 core plugs, 20 were selected for saturation study. These plugs were selected based on two criteria: 1) porosity of the plugs covers the entire porosity range; 2) measured results between CNG and GRI have obvious deviation.

We used Isopar-L, a mixture of branched saturated hydrocarbons (isoparaffins) primarily consisting of carbon chains with 11-13 carbon atoms, for saturation. It was selected for its relatively low viscosity (~1.5 cP at room condition). Prior to saturation, fluid preparation involved degassing the Isopar-L using a vacuum pump for approximately half an hour.

A 500 ml stainless steel accumulator served as a saturation cell. Core plugs were placed in the cell, closed in, then air was evacuated by vacuum. Afterward, Isopar-L was added into the accumulator until filling the vacuum, then pressure was gradually applied to 4000 psig. The plugs stayed at constant 4000 psig for one week. The pressure was then slowly relieved to retrieve plugs for measurement.

Before NMR measurement, sample surface fluid was removed by blotting with an Isopar-wetted towel. Weights were taken before running the samples in NMR.

NMR measurement. Fluid content within a plug was measured using a 12 MHz GeoSpec NMR spectrometer (Oxford Instruments) with a 53 mm probe. Data acquisition and processing was performed using software from Green Imaging (Green Imaging Technologies). The measurement is based on a CPMG pulse sequence (11, 12) with an echo time of 110 µs. The delay time between successive scans was 5 s. A total of 32 scans were accumulated for each sample. The absolute amount of fluid volume V_l in a sample was obtained by calibrating the measured NMR signal to a standard sample with a known amount of water.

Results and discussion

Porosity and density from CNG and GRI of all studied samples

We first examined the CNG results from the 224 core plugs. Fig. 2 shows the cross-plots of the porosity, bulk density, and grain density between CNG and GRI for the 224 samples. Bulk density, shown in Fig. 2b, exhibits a small systematic difference between CNG and GRI results. The reason is that GRI does not compensate for liquid lost from a sample between subsurface and ‘as received" states. In contrast, the CNG method measures empty space using gas-porosimetry and replaces it with light-hydrocarbon liquid as in Eq. (3). In view of this, Fig. 2 illustrates the measured results from CNG and GRI are largely consistent over the entire range of porosity and density for the 224 source rock samples.

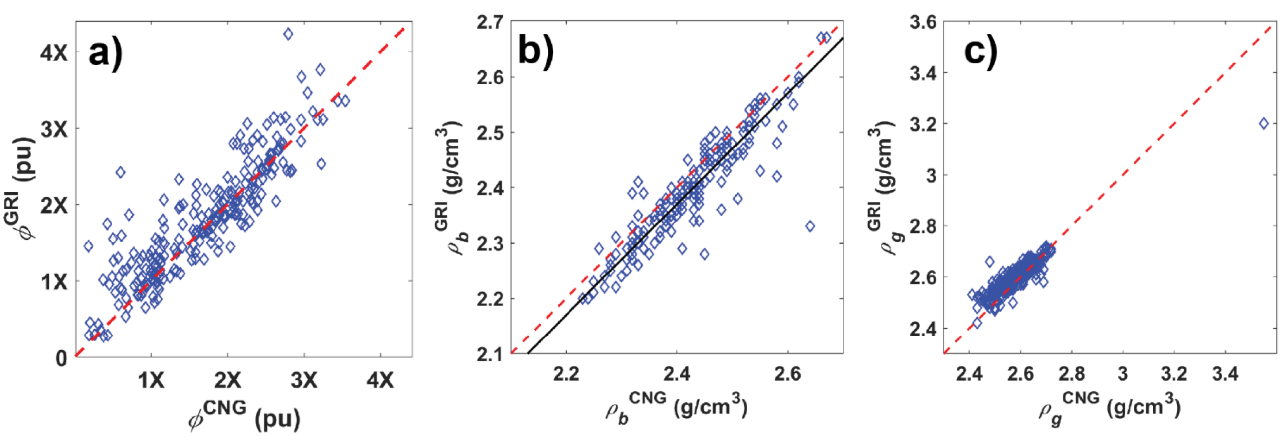

Figure 2—Cross-plots of measured (a) porosity, (b) bulk density, and (c) grain density from CNG and GRI for 224 samples from 19 wells. Red dashed lines are 1:1 line. The solid black line in (b) shows the fitting between the two methods.

It is evident that some samples deviated from the 1:1 line, as illustrated in Fig. 2. The primary reason for this deviation is the difference in sample sizes for CNG and GRI, as described in the test workflow and sampling sections. Specifically, CNG uses 1-inch diameter plugs, representing a 1-inch section of the whole core. Conversely, GRI samples are extracted from sections measuring 2 inches, and occasionally up to 6 inches in length, depending on the commercial laboratory's processing methods.

The studied reservoir exhibits significant heterogeneity at inch scale or smaller in many sections. The rock properties can vary dramatically within very small depth ranges, as evidenced by the examples showed in Fig. 3 and Fig. 4. Fig. 3a shows an image of a source rock whole core approximately 12 inches in length and 4 inches in diameter, with fluid content measured at each inch using high-spatial-resolution NMR (HSR-NMR) technology (*12*). The coloration of the core surface reflects the organic matter content within the rock, where darker shades represent higher organic matter content. Fluid content measured using HSR-NMR corelates with the organic matter content over the whole core with regions of higher organic matter containing more fluids. Clearly, Fig. 3a shows that organic matter and the enclosed fluid content varies at inch scale. Fig. 3b is a previously slabbed 2/3 core section carcass with visible color changes, suggesting TOC variation. Two plugs (plugs not shown) were drilled about 1 inch apart with a significant measured porosity difference between them: 11.7 pu versus 4.2 pu. The significant variation in rock properties over a small scale means that results from smaller samples, such as those used in the CNG method, can differ considerably from those obtained from larger samples, such as those used for the GRI method, even when both are taken centered around similar depths, as GRI represents the average property of a longer depth interval of whole core than that of CNG. For example, if the whole section of Fig. 3b was crushed for GRI, the expected GRI porosity is likely to be approximately 7.9 pu, very different from the plug porosity from CNG: 11.7 pu or 4.2 pu.

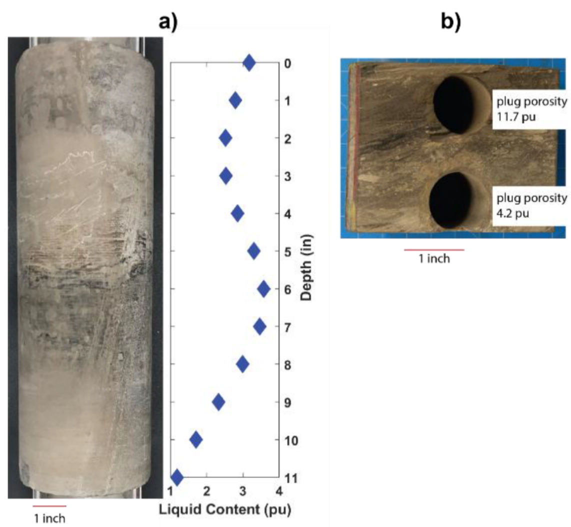

Figure 3—a) Photo and measured HSR-NMR fluid content of a whole core 12 inch in length and 4 inch in diameter; b) Photo of the 2/3 core section carcass where two plugs were drilled about 1 inch apart with measured porosity 11.7 and 4.2 pu, respectively.

Fig. 4 is another simple example illustrating that the sample size can result in significant differences in a measured property between CNG and GRI. One outlier sample in Fig. 2c shows unusually large grain density, 3.55 g/cm^3 from CNG and 3.20 g/cm^3 from GRI. The reason is that the rock contains a substantial amount of high-density mineral pyrite (5 g/cm^3), as illustrated in Fig. 4. The whole core in Fig. 4a again shows dramatic color change over a small depth scale. The plug for CNG in Fig. 4b clearly shows the presence of a large amount of pyrite with its typical golden color. The GRI sample, while also centered around the same depth, contains rocks from about a 2" section of the whole core. The saved 1/3 slab in Fig. 4c from the same section shows that pyrite remains abundant but is not homogeneously distributed, suggesting that the average pyrite content is lower in the GRI sample than in the CNG plug. Consequently, the grain density measured from the CNG plug is higher.

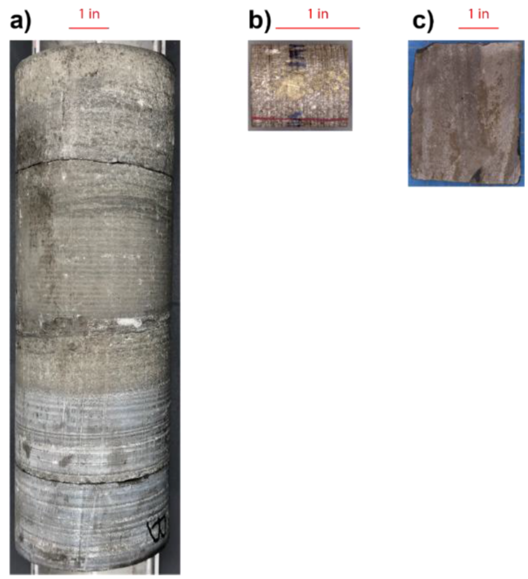

Figure 4—Images of the (a) whole core, (b) plug for CNG experiments, and (c) 1/3 slab. The plug (a) and slab (b) are both from the top section of the whole core. Note that (a) and (c) are on the same scale while (b) was enlarged to illustrate the pyrite in golden color.

While the trend of measured results between CNG and GRI is expected to be consistent for a large number of samples as shown in Fig. 2, comparing results on a single sample can lead to inaccuracies due to the different sample sets used by each method. Even if samples are taken from the same or adjacent rock sections, significant heterogeneity in many reservoirs can result in varying measured outcomes. If the results of CNG and GRI are similar or identical, it indicates that the studied reservoir section is homogeneous.

Porosity and density of saturated samples

The previous section shows that comparing the results of CNG and GRI directly can be misleading since they use samples of different sizes. In this section, we use the saturation method on a subset of plugs to evaluate CNG and GRI.

Fig. 5 shows the cross-plots of measured porosity and density between CNG and saturation of the selected 20 plugs under study. It is obvious the measured results from CNG and saturation are very consistent. For a few samples, porosity measured by CNG is slightly higher than that measured by the liquid saturation method, as pressure saturation at 4000 psig for one week may not completely saturate some the plugs. This is expected as source rocks have extremely small pore throats at the nm or even sub-nm scale. Some pores which may be accessible to helium gas used in gas porosimetry may not be easily penetrated by liquid, even for low viscosity Isopar-L. The challenge in fully saturating the core plugs using liquid demonstrates that CNG is an optimal method for characterizing source rocks. CNG employs inert gas like Helium to probe empty porosity in the plug, which is more effective than liquid used in our saturation experiment, because gas can easily penetrate all pores due to its much lower viscosity and smaller molecular size.

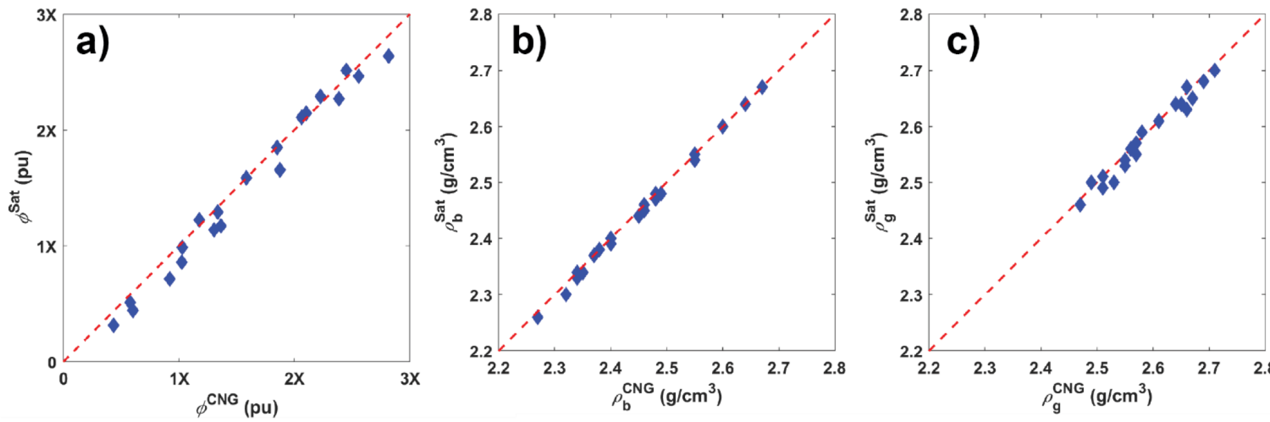

Figure 5—cross-plots of porosity (a), bulk density (b), and grain density (c) measured between CNG and saturation methods. Red-dashed lines are 1:1 line.

Fig. 6 shows cross-plots between GRI and saturation methods. Obviously, the scattering of the measured data between the two methods is much larger than those in Fig. 5. This again is due to the significant heterogeneity in the whole cores and differences in sample sizes and locations between the GRI samples and CNG plugs as discussed above.

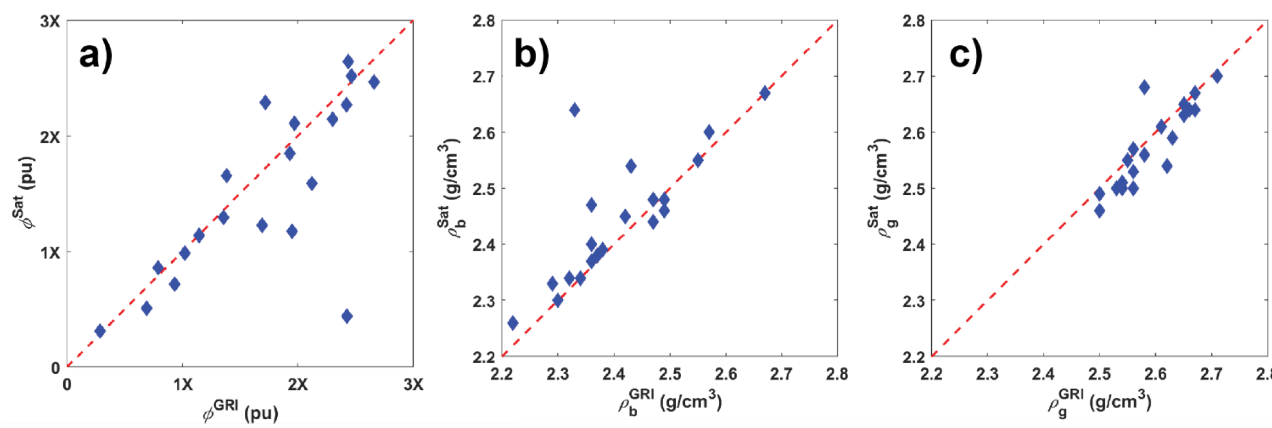

Figure 6—Cross-plots of (a) porosity, (b) bulk density, and (c) grain density measured between GRI and saturation methods. Red-dashed lines are 1:1 line.

The saturation method clearly shows that CNG provides consistent and accurate results. GRI may also provide correct results on the samples under investigation. But CNG holds substantial advantages over the GRI method:

- CNG is non-destructive and non-invasive, allowing the "as received" core plug to be used for further studies. In contrast, the GRI sample is crushed and subjected to solvent extraction/cleaning during the Dean-Stark distillation. The physical and chemical processes applied to the samples could alter the pore surface and connectivity in both organic and inorganic components in the source rocks.
- CNG is a rapid measurement method. It takes only an hour to complete the measurement of one sample. In contrast, GRI takes about a month to finish one sample and needs parallel setups for multiple samples.
- CNG requires only 10% of the sample mass (approximately 30 g) compared to GRI (approximately 300 g) and thus allows for better investigation of reservoir heterogeneity in thinly laminated formations.

- GRI requires crushed particles to be 20 – 35 mesh (0.5 – 0.84 mm). This size can cause biases since the yield depends on rock properties like TOC, clay content, and composition which may affect ease of crushing and the resulting particle sizes. Some constituents may disproportionately yield particles of usable sieve size while some lithologies will powder. Therefore, the measured result may not accurately reflect the average of the entire rock section.

Conclusion

The study evaluated the porosity and density measurement methods for organic-rich tight rocks, specifically comparing the combination of nuclear magnetic resonance and Gas-porosimetry (CNG) method with the industry-standard GRI method. The key findings are:

The porosity and density measurements from the CNG method were consistent with those from the saturation method, while significant scattering was observed between the CNG and GRI methods. The study concluded that results from different methods can be significantly different, due to variations in the samples used for different methods. The study highlighted the importance of considering formation heterogeneity when interpreting results from different methods. The significant variations in rock properties over small scales can lead to differences in measured outcomes between methods.

This study determines that the CNG method is reliable and optimal for determining petrophysical parameters of organic-rich source rock reservoirs. The CNG method is noninvasive, rapid, and can be completed in hours, whereas the GRI method is destructive and takes several weeks. The CNG method also requires only 10% of the sample mass compared to the GRI method, making it more efficient for investigating reservoir heterogeneity.

Overall, the study supports the broader application of the CNG method in the petroleum industry for characterizing unconventional source rock reservoirs.

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